

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 10: Circles (Core-Basics Edition)

CHAPTER 10: CIRCLES

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) CIRCLE KE PARTS

Centre = beech ka point. Radius (r) = centre se circle tak. Diameter = 2r. Chord = circle pe do points ko jodne wali line.

2) PERPENDICULAR (90 degree)

Do lines jo 90 degree pe milti hain. Symbol: chhota square corner pe.

3) RIGHT TRIANGLE + PYTHAGORAS

90 degree wale triangle mein:
 $hyp^2 = side1^2 + side2^2$.

4) SQUARE / SQUARE ROOT

$a^2 = a \times a$. $\sqrt{100} = 10$.
Tangent ki length nikaalne mein use hota.

5) TANGENT vs SECANT (is chapter ka dil)

SECANT = line jo circle ko 2 points pe CUT kare.
TANGENT = line jo circle ko 1 point pe TOUCH kare.
Touch point = "point of contact".

6) NUMBER OF TANGENTS

Andar ke point se: 0. Circle pe: 1. Bahar se: 2.

7) ANGLE BISECTOR

Ek line jo angle ko 2 barabar hisson mein baant-ti hai.

NOW: ASLI CHAPTER 10 SHURU

TOPIC 1: TANGENT TO A CIRCLE

Tangent = line jo circle ko sirf 1 point pe touch kare (cut nahi karti). Us point ko point of contact kehte.

TOPIC 2: THEOREM 1 (TANGENT-RADIUS)

EXAM ALERT: "Tangent radius ke perpendicular hota hai" - statement aur use har exam mein.

"Point of contact pe tangent, radius ke PERPENDICULAR (90 degree) hota hai."

[CORE] Isi se right triangle banta hai (radius + tangent + centre line) -> Pythagoras laga sakte ho.

TOPIC 3: THEOREM 2 (TWO TANGENTS)

EXAM ALERT: "Bahar ke point se 2 tangents barabar hote hain" - bahut questions ispe.

"External point se circle pe khinchi do tangents ki LENGTH barabar hoti hai ($PA = PB$)."

Extra: centre se point join karo -> wo angle APB ko bisect karta hai.

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Two tangents + angle

(Exam favourite)

P se do tangents PA, PB khinche. Angle APB = 60.
Prove triangle PAB equilateral (sab side barabar).

PA = PB (equal tangents) -> triangle isosceles.

Isosceles mein base angles barabar:

angle PAB = angle PBA.

Sum of angles = 180:

$60 + 2(\text{angle PAB}) = 180 \rightarrow \text{angle PAB} = 60.$

Saare angle 60 -> equilateral triangle. (Proved)

Solved Example 2 - Tangent length (Pythagoras)

(Exam favourite)

Radius 7 cm. Centre se 25 cm door point P. Tangent length PT?

Angle OTP = 90 (tangent perp radius).

$OP^2 = OT^2 + PT^2$

$25^2 = 7^2 + PT^2 \rightarrow PT^2 = 625 - 49 = 576$

PT = 24 cm. (7-24-25 triplet!)

Solved Example 3 - Angle between tangents

(Exam favourite)

Tangents 70 ka angle banate hain P pe. Radius aur tangent ke beech ka angle (centre wale quadrilateral se):

OAPB quadrilateral: angle OAP = angle OBP = 90.

Sum = 360 -> angle AOB = $360 - 90 - 90 - 70 = 110.$

Solved Example 4 - Equal tangents

P se PA aur PB tangents. PA = 10 cm. PB?

Equal tangents -> PB = PA = 10 cm.

Solved Example 5 - Tangent length (8-15-17)

Radius 8 cm, centre se 17 cm door. Tangent length?
 $PT^2 = 17^2 - 8^2 = 289 - 64 = 225 \rightarrow PT = 15 \text{ cm.}$

Solved Example 6 - Bisected angle

Tangents 80 ka angle banate P pe. angle OPA?
OP bisect karta hai $\rightarrow \text{angle OPA} = 80/2 = 40.$

Solved Example 7 - Tangent-radius angle

Tangent aur radius (point of contact pe) ke beech angle?
Hamesha 90 degree (Theorem 1).

Solved Example 8 (EASIEST) - Number of tangents

Circle ke bahar ke point se kitne tangents?
2 tangents (bahar ke point se hamesha 2).

Q AND A TIME

Q1. CORE: secant aur tangent mein fark ek line mein.

Q2. Radius 9 cm, centre se 15 cm door point. Tangent length nikaal. (Pythagoras)

Q3. P se PA, PB tangents. PA = 12 cm. PB?

Q4. Tangent aur radius ke beech (contact point pe) angle?

Q5. Tangents 90 ka angle P pe. angle OPA?

Q6. Circle ke andar ke point se kitne tangents khinch sakte hain?

Q7. CORE CHECK: 5-12-13 triplet use karke - radius 5, tangent 12 ho toh OP (centre se distance)?

SUMMARY

1. Tangent = 1 point pe touch; Secant = 2 points pe cut.
2. Theorem 1: tangent point of contact pe radius ke perpendicular (90°).
3. Theorem 2: external point se 2 tangents barabar ($PA = PB$).
4. Tangent problems mein right triangle banta hai -> Pythagoras.
5. Tangents count: inside 0, on circle 1, outside 2.

CORE (RED page) revise: circle parts, perpendicular, Pythagoras, tangent vs secant, number of tangents.